

PHARMACY CALCULATIONS & PHARMACOKINETICS

QUICK REFERENCE FORMULA SHEET – MCQ 2025

Highlighted = found on
PEBC formula sheet

SPECIFIC GRAVITY

= Weight of substance / weight of
equal volume of water

(for converting mass of liquid in volume and vice-versa)

CHEMISTRY

$$n \text{ (moles)} = \frac{m \text{ (mass)}}{M \text{ (molar mass)}}$$

To get...	Multiply n (moles) by
Mass	Molar mass
# of particles	6.02×10^{23}
Eq	Valence of ion
Osmols	# of dissociable ions or particles

Henderson-Hasselbalch:

$$pH = pK_a - \log \frac{[HA]}{[A^-]}$$

$$pOH = pK_b + \log \frac{[HB^+]}{[B]}$$

COMPOUNDING

% volume by volume (v/v)	mL per 100 mL
% weight by volume (w/v)	grams per 100 mL
% weight by weight (w/w)	grams per 100 grams

Dilution: $C_i V_i = C_f V_f$

Sensitivity = Maximum acceptable error (5%) x
minimum weighable quantity

Alligation

If 3 stock concentrations → systems of equations

If 2 stock concentrations → cross-subtract method

IBW: $X + (2.3 \text{ kg} \times \# \text{ of inches} > 5 \text{ ft})$ where **X = 45.5**
kg (female) or 50 kg (male)

BMI: weight (kg) / (height[m])²

CREATININE CLEARANCE

$$CrCl \text{ (mL/min, male)} = \frac{(140 - \text{age}) \times 1.2 \times \text{actual body weight (kg)}}{\text{serum creatinine } \left(\frac{\mu\text{mol}}{\text{L}} \right)}$$

$$CrCl \text{ (mL/min, female)} = 0.85 \times CrCl \text{ (male)}$$

CONVERSIONS

1 foot	12 inches
1 in	2.54 cm
1 tsp	5 mL
1 tbsp	15 mL
1 kg	2.2 lbs
1 mL	15-20 drops

Opioid Table "HOME"

Hydrocodone	6
Oxycodone	20
Morphine	30
codEine	200
25-50% reduction for incomplete cross tolerance (except morphine/fentanyl)	

**DON'T FORGET... WATCH
YOUR UNITS!!!**

ABSORPTION

Diffusion (Fick's Law) $\propto P_m, SA_m, \text{conc. Gradient}$

Dissolution (Noyes-Whitney) $\frac{D \times S}{V \times h} (C_s - C)$

$$\% \text{ ionization (weak acid)} = \frac{100}{1 + \text{antilog} (pK_a - pH)}$$

$$\% \text{ ionization (weak base)} = \frac{100}{1 + \text{antilog} (pH - pK_a)}$$

DISTRIBUTION: $V_d = \frac{\text{amount of drug in body (dose)}}{\text{plasma concentration (Cp)}}$

METABOLISM: Phase 1 (oxid., red., hydrolysis), Phase 2
(conjugation)

ELIMINATION

$$Cl = kV_d = \frac{F \cdot D}{AUC} = \frac{\text{rate of elimination}}{C_p} = Cl_r + Cl_h$$

$$Cl_r = \frac{\text{filtration} + \text{secretion} - \text{reabsorption}}{C_p}$$

$$Cl_h = \frac{Q \cdot f_u \cdot Cl_{int}}{Q + f_u \cdot Cl_{int}}$$

IV INFUSION

$$C_p = \frac{R_0}{Cl} \cdot (1 - e^{-kt})$$

$$Cl = K_e V_d$$

IV BOLUS, SINGLE DOSE

$$C_p = \frac{D}{V_d} \cdot e^{-kt} = C_{p0} \cdot e^{-kt}$$

IV BOLUS, MULTIPLE DOSE

$$C_{p_{\max,ss}} = \frac{F \cdot D}{V_d \cdot (1 - e^{-k\tau})}$$

$$C_{p_{\min,ss}} = C_{p_{\max,ss}} \cdot e^{-k\tau}$$

ORAL DRUG, SINGLE DOSE

$$C_p = \frac{F \cdot D \cdot k_a}{V_d \cdot (k_a - k)} \cdot (e^{-kt} - e^{-k_a t})$$

AVG. C_p

$$C_{p_{av,ss}} = \frac{AUC_0^\tau}{\tau} = \frac{F \cdot D}{Cl \cdot \tau} = \frac{F \cdot R_a}{Cl}$$

FLUCTUATION

$$\frac{C_{p_{\max,ss}}}{C_{p_{\min,ss}}} = \frac{1}{e^{-k\tau}}$$

ACCUMULATION

$$r = \frac{C_{p_{ss}}}{C_{p_1}} = \frac{1}{1 - e^{-k\tau}}$$

NON-LINEAR PK

$$F \cdot D = \frac{V_{\max} \cdot C_{p_{ss}}}{K_m + C_{p_{ss}}}$$

$$t_{1/2} = \frac{\ln(2)}{k}$$

LEGEND

P_m	Drug membrane permeability
SA_m	Membrane surface area
D	Drug diffusion coefficient (in Noyes-Whitney) OR Dose
S	Drug surface area
V	Volume of fluid
H	Diffusion layer thickness
C_s	Drug solubility
K	Elimination rate constant
F	Bioavailability
AUC	Area under curve
Cl_r	Renal clearance
Cl_h	Hepatic clearance
Q	Hepatic blood flow
f_u	Unbound drug fraction
Cl_{int}	Intrinsic clearance
C_p	Plasma concentration
R_0	Infusion rate
t	time
C_{p0}	Plasma concentration at time zero
SS	Steady-state
τ	Dosing interval
R_a	Average rate of drug administration
C_{p1}	Plasma concentration after one dose